

Formalizing Entropic Time in the Osacra-Gravaser Framework via Homotopy Type Theory

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Abstract

Recent experimental confirmations using isolated Bose-Einstein Condensates (BECs) have demonstrated that time can be dynamically parameterized as internal entropy exchange. This paper formalizes this emergent, vectorized entropic time within the Osacra-Gravaser framework using the rigorous foundations of Homotopy Type Theory (HoTT). By mapping the 14-dimensional fractal manifold topologies to types and paths, we prove that the accumulation of discrete thermodynamic polarization layers natively constructs the arrow of time. This provides a direct path for computational verification in proof assistants such as LEAN.

1 Introduction

In the Osacra-Gravaser framework, gravity and spacetime are not fundamental geometric continuums but emergent thermodynamic manifestations. We postulate that the macroscopic event horizon is structured as a discrete fractal manifold (Hausdorff dimension $D_H = 3/2$), governed by the fractional coherent equation:

$$l_p^{1/2} \square^{3/4} h_{\mu\nu}^{(1)} + R_{\mu\nu}^{(2)} = 0 \quad (1)$$

Experimental physics has recently confirmed via BECs that quantum evolution can be governed strictly by internal coarse-grained entropy exchange, creating an “entropic time.” To elevate this from a physical framework to a mathematically rigorous, verifiable theorem, we deploy Homotopy Type Theory. In HoTT, spaces are treated as types, and continuous deformations (time evolution) are treated as paths (identities) between these types.

2 Type-Theoretic Foundations of the Fractal Horizon

Let $\mathcal{M}^{14} : \text{Type}$ define the fundamental 14-dimensional phase space of the framework. We define a discrete polarization layer as a dependent type over the manifold:

$$\text{Layer} : \mathcal{M}^{14} \rightarrow \text{Type} \quad (2)$$

Each frozen layer n contributes precisely two units of von Neumann entropy. We define the entropy function as a mapping from a layer to the real numbers (or a suitable discrete semiring in LEAN):

$$\text{Ent} : \prod_{x:\mathcal{M}^{14}} \text{Layer}(x) \rightarrow \mathbb{R} \quad (3)$$

such that for any inhabitant $l : \text{Layer}(x)$, $\text{Ent}(l) = 2 \ln 2$.

3 Time as Path Concatenation (Vectorized Entropy)

In standard physics, time is an external parameter t . In our HoTT formalization, time is an emergent path connecting distinct thermodynamic states.

Definition 1 (Entropic Path). *Let $S_A, S_B : \text{Type}$ be two thermodynamic microstates of the condensate. An entropic time vector $\vec{\tau}$ is a path $p : S_A =_{\text{Type}} S_B$ induced by the univalence axiom from an equivalence $e : S_A \simeq S_B$ that carries a non-zero entropy gradient ΔS .*

By the Univalence Axiom, $(S_A = S_B) \simeq (S_A \simeq S_B)$. The continuous transport of a quantum state ψ through time is simply the transport along this path:

$$\text{transport}^P(p, \psi_A) = \psi_B \quad (4)$$

where P is the family of ghost condensate wavefunctions.

Theorem 1 (Arrow of Time via Path Concatenation). *The macroscopic time parameter $t(N)$ derived from the accumulation of N layers is isomorphic to the iterated concatenation of paths in the universe $\mathcal{U}$.*

Proof. Let $l_1, l_2, \dots, l_N$ be a sequence of polarization layers. The transition between layers is a thermodynamic bridge $\delta Q = T\delta S$, corresponding to a path $p_n : l_{n-1} = l_n$. The total temporal evolution from the initial state l_0 to l_N is the concatenated path:

$$p_{\text{total}} = p_1 \cdot p_2 \cdot p_3 \cdots p_N \quad (5)$$

The length of this path, mapped via the **ap** (action on paths) of the Entropy functor, yields the exact time accumulation equation from the framework:

$$\text{ap}_{\text{Ent}}(p_{\text{total}}) = \sum_{n=1}^N \text{Ent}(l_n) = N(2 \ln 2) \quad (6)$$

Because path concatenation is associative but fundamentally directed when mapped to the non-invertible entropy functor (due to the Second Law implemented as a directed type or category constraint), the irreversibility of this sum formally defines the Arrow of Time. $\square$

4 Topological Matter and Winding Numbers

Matter manifests as topological knots piercing this stacked entanglement. In HoTT, this corresponds to the higher inductive properties of $\mathcal{M}^{14}$, specifically its fundamental group $\pi_1(\mathcal{M}^{14})$. The particle spin J , defined previously as $J = \frac{\hbar}{2\pi} \sum w_n$, is formalized as the loop space $\Omega(\mathcal{M}^{14}, x_0)$. The winding number w_n is the degree of the map $S^1 \rightarrow \mathcal{M}^{14}$, which naturally quantizes because $\pi_1(S^1) = \mathbb{Z}$.

5 Conclusion and LEAN Implementation Strategy

By translating the fractional coherent equation and the discrete entropy accumulation into Homotopy Type Theory, we have established a foundation where “vectorized entropy” is rigorously defined as path concatenation in the universe of types. The next step is to encode $\mathcal{M}^{14}$, the univalence-derived entropic paths, and the Fractional Finite Difference Operator $\Delta^{3/4}$ directly into LEAN to automate the verification of the backreaction mechanics.